CREATING VISUALS IN POWER BI

Power BI is an interactive business intelligence and data visualization tool that allows users to connect to a wide variety of data sources, prepare and model data, and finally represent it through rich visuals. Data visualization is a core component of Power BI, and understanding how to create, customize, and manage different visual types is essential for building insightful reports. This detailed note explores the complete workflow of importing data for visuals, understanding the visualization pane, and creating and formatting visuals—centered around the example of creating a stacked bar chart.

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1. Starting a New Power BI Report and Importing Data

The first step in building any visualization using Power BI is creating a new report and importing a dataset. In this example, we begin by opening Power BI Desktop and clicking on **Blank Report**. A blank report canvas is loaded, and we can connect to a data source by using:

Home → Get Data → Excel

We select an Excel file named **Visuals.xlsx**, which contains sample data intended for building different types of visuals. After selecting the file, Power BI shows a navigator window that lists available sheets. Here, we select **Sheet1** (Sheet number one) and check its preview.

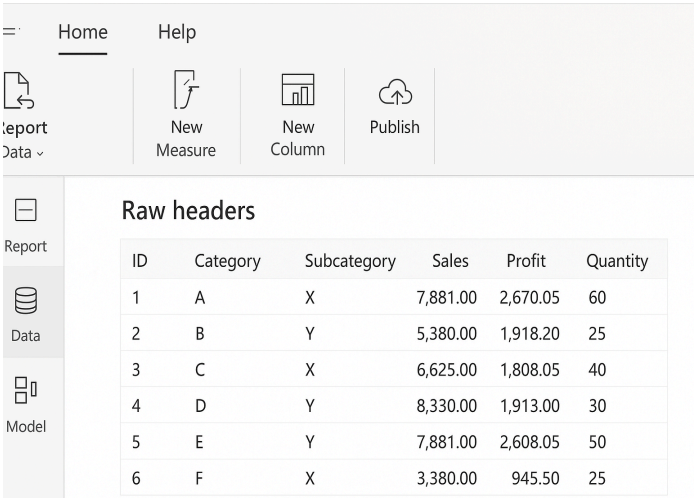
Before loading the dataset, it is recommended to inspect and clean the data using **Power Query Editor**. This is done using:

Transform Data → Power Query Editor

Inside the Power Query Editor window, we review the following aspects:

1.1 Row Headers

Power BI automatically detected the first row as headers. Since headers are correctly identified, there is no need to use the “Use First Row as Headers” option.



ID	Category	Subcategory	Sales	Profit	Quantity
1	A	X	7,881.00	2,670.05	60
2	B	Y	5,380.00	1,918.20	25
3	C	X	6,625.00	1,808.05	40
4	D	Y	8,330.00	1,913.00	30
5	E	Y	7,881.00	2,608.05	50
6	F	X	3,380.00	945.50	25

1.2 Data Types

Each column's datatype must be validated to ensure correct usage in visuals:

- **ID column**: Though numeric in appearance, IDs represent categorical information. Therefore, instead of keeping it as a number, we convert it to **Text**.
  This ensures that Power BI doesn't treat it as a measure or quantity.

  A new step is created using **Add Step**, rather than replacing the existing datatype transformation.

- **Category column**: Recognized as **Text** (ABC). Correct.
- **Subcategory column**: Recognized as **Text** (ABC). Correct.
- **Sales, Profit**: Recognized as **Decimal Numbers (1.2)** – correct.
- **Quantity**: Recognized as **Whole Number (1)** – correct.
- **Date**: Properly identified as **Date data type**.
- **Region**: Categorical (Text) – appropriate.
- **Age**: Numeric – correct.
- **Satisfaction**: Numeric scale (1–10), correctly identified as a whole number.
- **CON column**: Documented as Decimal Number (1.2), correct.
- **Return Customer**: Yes/No categorical – correctly recognized as text.

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Except for the ID column, all other datatypes were correct.

After reviewing the dataset, we click **Close & Apply**. Power BI loads the data into the report model.

1.3 Renaming the Table

The table name defaults to the sheet name (Sheet1). For clarity, renaming the table is recommended:

- Right-click the table → **Rename**
- Rename to **VisualsData**

This practice improves report readability.

2. Introduction to the Visualization Pane

Power BI features a rich collection of visualization templates accessible from the **Visualizations Pane**. The pane includes icons representing different visuals such as bar charts, column charts, line charts, area visuals, maps, KPI cards, tables, matrix, decomposition tree, funnel, scatter, donut, treemap, and more.

2.1 Available Visual Categories

A quick overview:

- **Bar and Column Charts**
    - Stacked Bar
    - Stacked Column
    - Clustered Bar
    - Clustered Column
    - 100% Stacked Bar
    - 100% Stacked Column
- **Line and Area Charts**
    - Line Chart
    - Area Chart
    - Stacked Area Chart
    - 100% Stacked Area Chart
    - Combo Charts (Line + Column)
- **Advanced and Analytical Charts**
    - Ribbon Chart
    - Waterfall Chart
    - Funnel Chart
    - Scatter Chart

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- **Pie-Based Charts**
    - Pie Chart
    - Donut Chart
- **Maps**
    - Treemap
    - Filled Map
    - Azure Map
    - ArcGIS Maps
- **Cards and KPIs**
    - Card
    - Card (New)
    - KPI
- **Tables and Matrix Visuals**
    - Table
    - Matrix
- **AI Visuals**
    - Key Influencers
    - Decomposition Tree
    - Q&A
- **Other Tools**
    - Paginated Report
    - Slicers (Text, Dropdown, List, Button)
    - Power Apps visual
    - Power Automate visual

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2.2 Getting More Visuals

Users can import additional custom visuals.

- Click the three dots (…) under the Visualizations pane.
- Choose **Get More Visuals**
- Opens Microsoft AppSource with free/paid visual add-ons.
- Also possible to import visuals from a .pbiviz file.

This provides extendibility beyond Power BI's default library.

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3. Methods of Creating Visuals

There are two primary ways to create visuals in Power BI:

Method 1: Selecting a Visual Template First

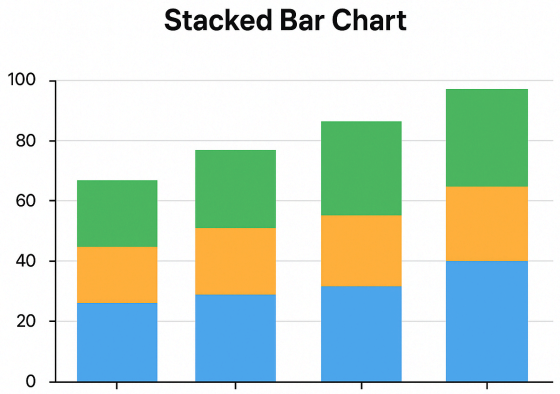
- Click an icon in the Visualizations pane.
- A blank visual placeholder appears on the canvas.
- Add data fields by:
    - Dragging fields into the Fields well, or
    - Clicking checkboxes in the Fields list (adds them automatically).

Method 2: Selecting Fields First

- Check the box next to a field in the Fields pane.
- Power BI automatically decides the most appropriate visual based on the field type.
- You can then change the visual type manually if needed.

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4. Creating a Stacked Bar Chart



A stacked bar chart is used to represent data using:

- At least **one categorical field** (for grouping)
- At least **one numerical measure** (to represent bar length)
- Optionally, another categorical field for stacking layers.

4.1 Building the Visual

1. Click on the **Stacked Bar Chart** template.
2. A blank visual appears with the prompt:
   "Select or drag fields to populate the visual."
3. Add fields:
   - **Category → Y-axis**
   - **Profit → X-axis**

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Power BI displays five bars representing categories A, B, C, D, and E.

4.2 Adding Stacked Layers (Optional)

To convert this into a true stacked bar chart:

- Drag **Subcategory** to **Legend**.

Now the bars display segments (X, Y, Z) stacked within each category, each shaded distinctly.

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5. Formatting the Visual (Format Visuals Section)

Power BI provides rich formatting options categorized into:

1. **Visual Formatting**
2. **General Formatting**
3. **Analytics Formatting**

Below is a detailed exploration of the formatting features.

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5.1 Format Your Visual – Detailed Breakdown

This section determines the appearance of axes, titles, labels, gridlines, colors, and more.

5.1.1 Y-Axis Formatting

Options include:

- **Font size**: Increase for readability (e.g., 18 px)
- **Axis color**: Change to dark black for clarity.
- **Axis position**: Left or right.
- **Concatenate labels**: Useful when hierarchical categories exist.
- **Axis title**:
    - Default shows the category field.
    - Font size, style, alignment, underline options available.

5.1.2 X-Axis Formatting

Options similar to Y-axis:

- Change font size
- Adjust color
- Show/hide axis
- Set decimal places
- Adjust display units (auto, thousands, millions)
- Customize axis title (e.g., “Profit”)

5.1.3 Gridlines

- Can be switched on/off.
- Useful for numeric comparison.
- Gridline color, style, and transparency options available.

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5.1.4 Zoom Slider

Enables horizontal zooming:

- Allows focused inspection of data.
- Particularly useful when there are many categories.

5.1.5 Bars (Data Colors)

You can:

- Change bar fill color
- Add border around each bar
- Adjust transparency
- Modify inner padding (space between bars)
- Enable rounded bar edges (optional; may distort aesthetics)

5.1.6 Data Labels

Data labels show exact values on bars.

Settings:

- Label position:
    - Auto
    - Outside end
    - Inside center
    - Inside base
    - Inside end
- Change label:
    - Font size
    - Color (white works best on dark bars)
- Background options:
    - Add background color behind label
    - Control transparency

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5.1.7 Plot Area Background

Allows adding:

- A solid background color
- An image (logo/branding)
- Fit/stretch/crop options depending on image.

5.2 Legend Formatting

The legend appears when a stacking category is added.

Options include:

- **Position** (top, bottom, left, right, centered)
- **Font styling and size**
- **Legend title** (enable/disable)
- **Legend symbol customization**

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5.3 General Formatting

This impacts the visual container rather than data representation.

Settings include:

- **Visual size (height/width)**
- **Padding**
- **X/Y positioning**
- **Lock aspect ratio**
- **Turn on/off header icons**

5.4 Title Formatting

The title of the visual is crucial for readability.

Options include:

- Custom title text (e.g., *Category-wise Profit Distribution*)
- Font size and style (bold/underline)
- Alignment (center, left, right)
- Background color
- Transparency control

You can also add a **subtitle** with its own formatting settings.

5.5 Border and Effects

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You can add:

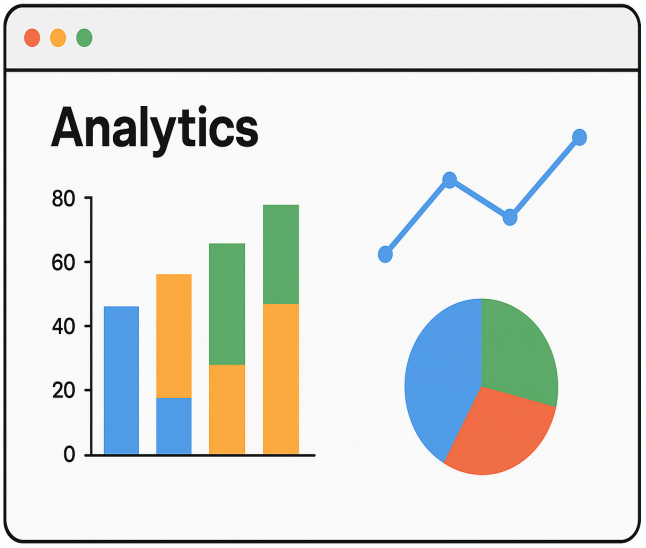
- Border around the visual
- Rounded corners (radius adjustment)
- Shadow effects
- Background color for entire visual box

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6. Analytics Pane (Advanced Analytics)

The analytics pane offers advanced features depending on the visual type.

In a stacked bar chart, you can:



6.1 Add Constant Line

Example:
Add a constant value of **1000** to mark a benchmark profit.

Options include:

- Line style
- Color
- Transparency
- Data label for the constant line

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Depending on the visualization type, analytics pane may also support:

- Trend lines
- Forecasting
- Min/Max lines
- Percentile lines
- Error bars
- Symmetry shading

However, these options vary across visuals and are not universally available.

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7. Summary of the Stacked Bar Chart Creation Process

Step-by-Step Summary

1. Open Power BI → Create a new report.
2. Import Excel file → Select relevant sheet.
3. Enter Power Query Editor for datatype validation.
4. Apply transformations → Close & Apply.
5. Insert stacked bar chart visual.
6. Add fields:
    - Category → Y-axis
    - Profit → X-axis
    - Subcategory → Legend (optional).
7. Format axes, gridlines, labels, colors, and legend.
8. Modify title, fonts, and layout.
9. Add analytical elements such as constant lines.

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10. The final output is a professionally formatted stacked bar chart.

8. Importance of Visual Formatting in Power BI

Proper visual formatting ensures:

- Professional presentation
- Easier interpretation of insights
- Improved readability
- Consistency across dashboards
- Effective highlighting of key metrics

A well-designed visual significantly enhances the storytelling aspect of data analysis.

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9. Additional Notes on Best Practices

9.1 Choosing the Right Visual

- Use bar/column charts for categorical comparisons.
- Use line charts for trends over time.
- Use pie/donut charts sparingly.
- Use treemap for hierarchical part-to-whole relationships.
- Use scatter plots for correlation analysis.

9.2 Avoid Chart Junk

- Minimize unnecessary colors.
- Avoid over-styling.
- Keep labels readable and uncluttered.

9.3 Use Consistent Formatting

- Maintain uniform font sizes across visuals.
- Use one color theme.
- Keep legends clear and concise.

9.4 Use Slicers for Interactivity

Slicers allow users to filter visuals easily, improving report usability.

10.Conclusion

Creating visuals in Power BI involves more than simply placing fields into charts. It is a complete workflow consisting of data import, cleaning, selecting visualization types, assigning fields, and thoroughly formatting the visual for clarity and professionalism. The stacked bar chart example demonstrates how Power BI supports both simple and advanced customization to deliver meaningful and insightful dashboards. With proper usage of axes, labels, legends, colors, and analytics, Power BI visuals can become powerful tools for data-driven decision-making.

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